Predicting Student Dropout in Higher Education Using Supervised Machine Learning Classification

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Background of the Problem

Student dropout is one of the most persistent challenges facing higher education institutions worldwide, and Kenya is no exception. When a student leaves university before completing their program, the student loses the time, money and opportunities invested in their education, the institution loses tuition revenue and falls short of its graduation targets, and the country loses skilled graduates it needs for economic growth. The scale can be large: in the dataset used in this project, 1,421 of 4,424 students (32%) dropped out. Dropout is rarely sudden. It is usually preceded by warning signs such as poor early academic performance, unpaid tuition, financial hardship, and gaps in family or social support, but these signs are rarely examined together until it is too late to help.

The problem this project addresses is therefore the late identification of at-risk students. Most institutions identify them manually or only after they have stopped attending, so support such as counselling, bursaries and academic mentoring arrives too late. A data mining model is the most viable solution for three reasons. First, it learns directly from historical student records that the institution already holds, so no new data collection is needed. Second, a supervised classification model can screen thousands of students every semester at almost no extra cost, which manual monitoring cannot match. Third, it shows which factors matter most, such as first-semester results or tuition status, so limited resources can be targeted at the students who need them most and decisions rest on evidence rather than guesswork.

Methodology

The main objective of this project is to build a classification model that accurately predicts whether a student will drop out or not, based on demographic, socio-economic and academic data, so that institutions can identify at-risk students early. Because the outcome we want to predict is a category and not a number, this is a classification task, a supervised data mining category in which the model learns from historical records with known outcomes and then predicts the outcome for new students. We chose classification over regression, which predicts numbers, and over clustering or anomaly detection, which work without known outcomes, because the institution already holds labelled outcomes and needs a clear flag (at risk or not) to act on. We used the "Predict Students' Dropout and Academic Success" dataset from the UCI Machine Learning Repository (Realinho et al., 2021), which contains 4,424 student records and 34 features, including admission grades, tuition payment status, age at enrollment, and first- and second-semester academic performance. The original three outcomes were merged into two, Dropout (1,421 students) and Non-Dropout (3,003 students, combining Enrolled and Graduate), because our aim is to flag dropout risk and the Enrolled group overlaps heavily with the other two.

The model was built in Python (pandas, scikit-learn, imbalanced-learn and XGBoost) in six steps. First, the data was cleaned by checking for missing values and duplicates (none were found); the categorical fields, such as course and application mode, were already numerically coded, so no further encoding

was needed. Second, the data was split 80% for training (3,539 students) and 20% for testing (885 students) using stratified sampling, so that both sets keep the same class proportions. Third, the features were scaled so that variables with large ranges, such as grades, do not dominate the others. Fourth, because Dropout is the minority class (about 32%), SMOTE was applied to the training data only, to balance the classes without leaking information from the test set. Fifth, four supervised algorithms were trained: Decision Tree (easy to interpret), Logistic Regression (a simple baseline), and Random Forest and XGBoost (ensemble methods that combine many trees and usually achieve higher accuracy). Sixth, the models were validated using 10-fold stratified cross-validation on the training set and the held-out test set, and compared using accuracy, precision, recall, F1-score and a confusion matrix, with the best model selected by its F1-score on the Dropout class.

Results and Discussion

Four supervised classifiers were trained on 3,539 student records and evaluated on a held-out test set of 885 students. XGBoost performed best, achieving a test accuracy of 89.0% and a 10-fold cross-validation accuracy of 87.6%. For the Dropout class it reached a precision of 85.3%, a recall of 79.6% and an F1score of 82.3%. Random Forest followed closely with a test accuracy of 88.8% and an F1-score of 82.0%, then Logistic Regression (87.5% accuracy) and the Decision Tree (85.8%). The small gap between crossvalidation and test accuracy shows that the models generalize well and are not overfitted. Logistic Regression had the highest dropout recall (84.2%), meaning it caught the most at-risk students, but at a lower precision (78.4%), so it raised more false alarms.

Model	CV accuracy (10fold)	Test accuracy	Precision	Recall	F1-score
Decision Tree	84.6%	85.8%	79.3%	75.4%	77.3%
Random Forest	86.9%	88.8%	84.6%	79.6%	82.0%
Logistic Regression	86.0%	87.5%	78.4%	84.2%	81.2%
XGBoost	87.6%	89.0%	85.3%	79.6%	82.3%

Precision, recall and F1-score are for the Dropout class.

The project's objective was to build a model that predicts student dropout reliably enough to support early intervention, and this objective was met: all four models exceeded 85% test accuracy, and the two ensemble models reached about 89%. Feature importance analysis showed that the number and grades of approved curricular units in the first and second semesters were the strongest predictors, followed by tuition fee status, age at enrollment and course of study. This confirms that early academic performance and financial standing are the main warning signs of dropout. A limitation is that about one in five dropouts was still missed by the best model, so it should support, not replace, the judgment of student support staff.

Summary and Conclusion

This project developed a supervised data mining model to predict student dropout in higher education, addressing the problem that at-risk students are usually identified too late. Using the UCI "Predict Students' Dropout and Academic Success" dataset of 4,424 students, we merged the outcomes into Dropout and Non-Dropout, scaled the data, balanced the classes with SMOTE, and trained Decision Tree, Random Forest, Logistic Regression and XGBoost classifiers, validated with a stratified 80/20 split and 10-fold cross-validation. XGBoost was the best model, with a test accuracy of 89.0% and a dropout F1-score of 82.3%, showing that institutional records can identify students at risk of leaving. Future work should include hyperparameter tuning, collecting more local data from Kenyan institutions, adding features such as class attendance and family support, predicting the Enrolled class separately, and deploying the model in a simple dashboard so administrators can screen students each semester and offer counselling, bursaries and mentoring early.

Project repository: https://github.com/MarkLemute/student-dropout-prediction

Reference

Realinho, V., Machado, J., Baptista, L., & Martins, M. V. (2021). *Predict Students' Dropout and Academic Success.* UCI Machine Learning Repository. https://archive.ics.uci.edu/dataset/697